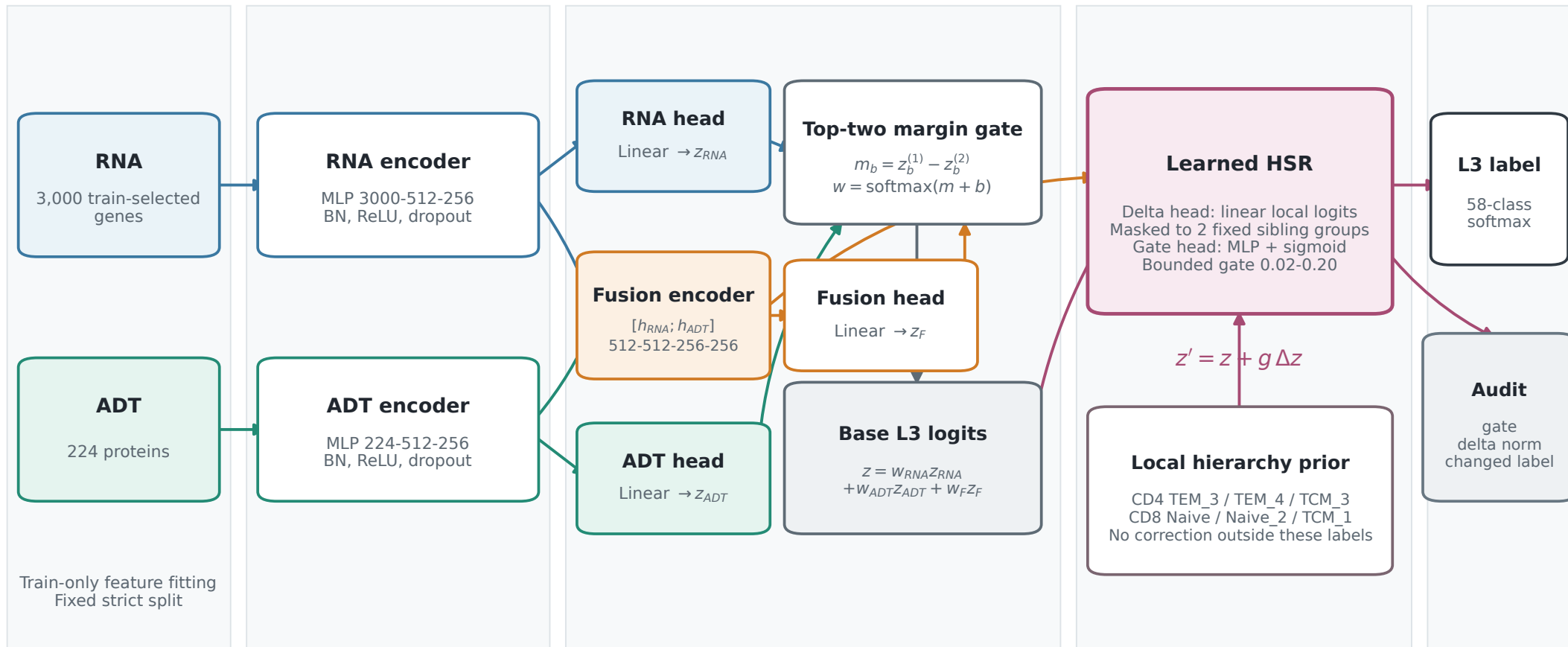


a CITE-seq input**b Modality encoders****c Margin-gated base classifier****d Learned HSR****e Outputs**

Training: final CE + 0.35 branch CE + 0.50 fusion CE + 0.05 teacher distillation + 0.20 HSR auxiliary loss; modality dropout = 0.15